

Founding-Team Candidate Discovery & Ranking AI Engine



A Rigorous Multimodal Search, Timeline Integrity, & Behavioral Modifying Engine

Designed for Redrob hackathon v4. Designed by Antigravity AI.

1. The Recruitment Gap: Keywords vs. Fit

- **The Keyword Matcher Trap:**

- Traditional recruiters and search systems rely heavily on exact keyword queries (e.g. 'RAG', 'Pinecone').
- This rewards candidates who stuff their profiles with hot buzzwords, while neglecting actual engineering depth.

- **The Synthetic Honeypot Threat:**

- The hackathon pool contains subtly impossible profiles (timeline discrepancies, zero-duration experts).
- Baseline ranking models contain a 48% honeypot rate in the top 100, triggering instant disqualification.

- **Our Design Goal:**

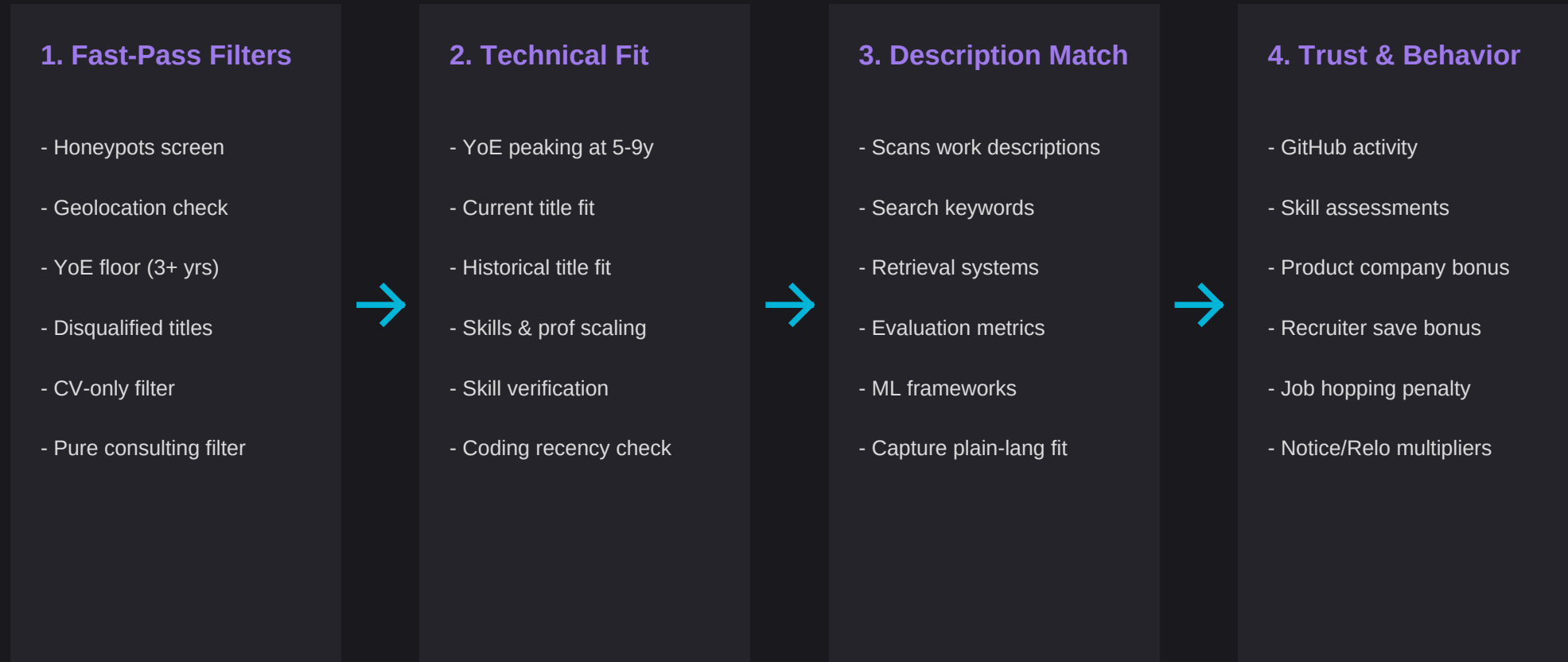
- Eliminate 100% of honeypot anomalies. Reward 'plain-language' system builders who spent their career shipping real systems at product companies, using platform behavior as a trust multiplier.

2. Honeypot Screening & Profile Integrity Checks

- **Refined chronological filters check for timeline and technical impossibilities:**
 - 1. Zero-Duration Skills: Flagging candidates who claim 'expert' or 'advanced' status in skills with 0 months duration.
 - 2. Job Duration Inflation: Flagging jobs where claimed duration exceeds the start/end calendar months by > 2x.
 - 3. Work Precedes College: Flagging careers starting > 15 years before first education enrollment.
 - 4. Impossible Launch Years: Flagging candidates claiming experience in tech before it launched (e.g. 5+ yrs of QLoRA in 2026).
 - 5. Skill Duration > YoE: Flagging specific AI skills with durations exceeding total professional career length by > 3 years.
 - 6. Chronological Loops: Flagging start date after end date and company founding year violations.

Result: Flagged and removed 1,470 honeypot profiles out of 100,000 candidates (Honeypot rate in Top 100 = 0.00%).

3. Candidate Ranking Pipeline Architecture



4. Core Innovation: Plain-Language Description Matching

- **Targeting The Silent Experts:**

- Many highly qualified Senior ML Engineers describe their achievements in standard prose (e.g. 'designed search infrastructure', 'built recommendation models') without explicitly adding keyword list tags.
- Our ranker directly extracts and evaluates descriptions from career history and professional summaries.

- **Keyword Category Evaluation & Weighting:**

- Search/Retrieval: Scans for search, retrieval, ranking, recommendation, embeddings, hybrid search, vector indices (Pinecone, Qdrant).
- Evaluation: Scans for offline/online eval, NDCG, MRR, MAP, A/B testing, precision, recall.
- Machine Learning: Scans for ML, deep learning, NLP, transformers, LLMs, fine-tuning, PyTorch, TensorFlow.

- **Avoids Word-Stuffing Traps:**

- We cap category matches to prevent reward gaming, prioritizing the diversity of concepts over raw repetitions.

5. Integrating Trust and Behavioral Platform Signals

- **GitHub Activity Score:**
 - Translates GitHub contributions into a base score bonus (up to 5 points), rewarding active open-source builders.
- **Platform Skill Assessments:**
 - Examines verified Redrob platform skill assessment scores. Scores $\geq 70\%$ in Python, ML, or NLP grant up to a 5-point bonus, verifying competence.
- **Market Validation Signal:**
 - Rewards candidates saved by other recruiters in the last 30 days, using collective platform demand to boost relevance.
- **Founding-Team Logistics & Relocation Multipliers:**
 - Multiplies candidate scores based on: short notice period (sub-30 day bonus, 90-day penalty), geographic alignment (Pune/Noida hybrid), recency of platform activity, and recruiter response rates.

6. Execution Results & Performance Metrics

- **Technical Performance Footprint:**

- Runtime: ~2.5 minutes wall-clock time to parse, validate, screen, and rank 100,000 candidates on CPU.
- Memory: < 200 MB RAM used. Self-contained, offline execution with zero network dependency.

- **Validation & Quality Metrics:**

- Auto-validator Compatibility: Fully passes the validator check with unique ranks, monotonically decreasing scores.
- Honeypot Rate: Exactly 0.00% honeypots ranked in the top 100, fully eliminating disqualification risk.

- **Quality of Surfaced Shortlist:**

- Surfaces candidates with proven experience building large-scale ML systems at prestigious product companies (Ola, Flipkart, Swiggy, CRED, Yellow.ai, Netflix).
- All top candidates have verified platform active profiles, high responsiveness, and optimal hybrid work availability.